

Total No. of Questions : 8]

Total No. of Printed Pages : 2

Roll No

AU/ME-4005-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Machine Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat sketch, if required.

1. a) State the maximum shear strain theory and draw its graph. 7
b) Define stress concentration. What are the causes of stress concentration? 7
2. a) What are different modes of failure? Explain. 7
b) What do you mean by Fatigue? 7
3. a) Discuss the modified Goodman diagram's uses. 7
b) Discuss the role of Fasteners and Keys. 7
4. a) Discuss the power screws use in brief. 7
b) The shear strength, tensile strength & compressive strength of a rivet joint are 100N, 120N and 150N respectively. If strength of unriveted plate is 200N. Find the efficiency of rivet joint. 7

AU/ME-4005-CBGS

PTO

[2]

5. Derive an expression for equivalent stiffness of two springs connected, having stiffness K_1 and K_2 . 14
 - i) In series
 - ii) In parallel
6. a) Write design steps for bolted joints. 7
b) What do you mean by stable and unstable operation? 7
7. a) What are the criteria for selection of ball and roller bearings? 7
b) Write short note on boundary lubrication. 7
8. a) Differentiate fillet and butt welds with neat sketches. 7
b) Write short notes on any two of the following. 7
 - i) Cyclic loading
 - ii) Surface factor
 - iii) Size factor
 - iv) Loading factor
